

1 November 2025, 제주

한 국 우 주 과 학 회 가 을 학 술 대 회

***Prediction of
Heavy Ion Charge State Ratios
& Elemental Composition of Solar Wind
Using Deep Learning***

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1. 서론

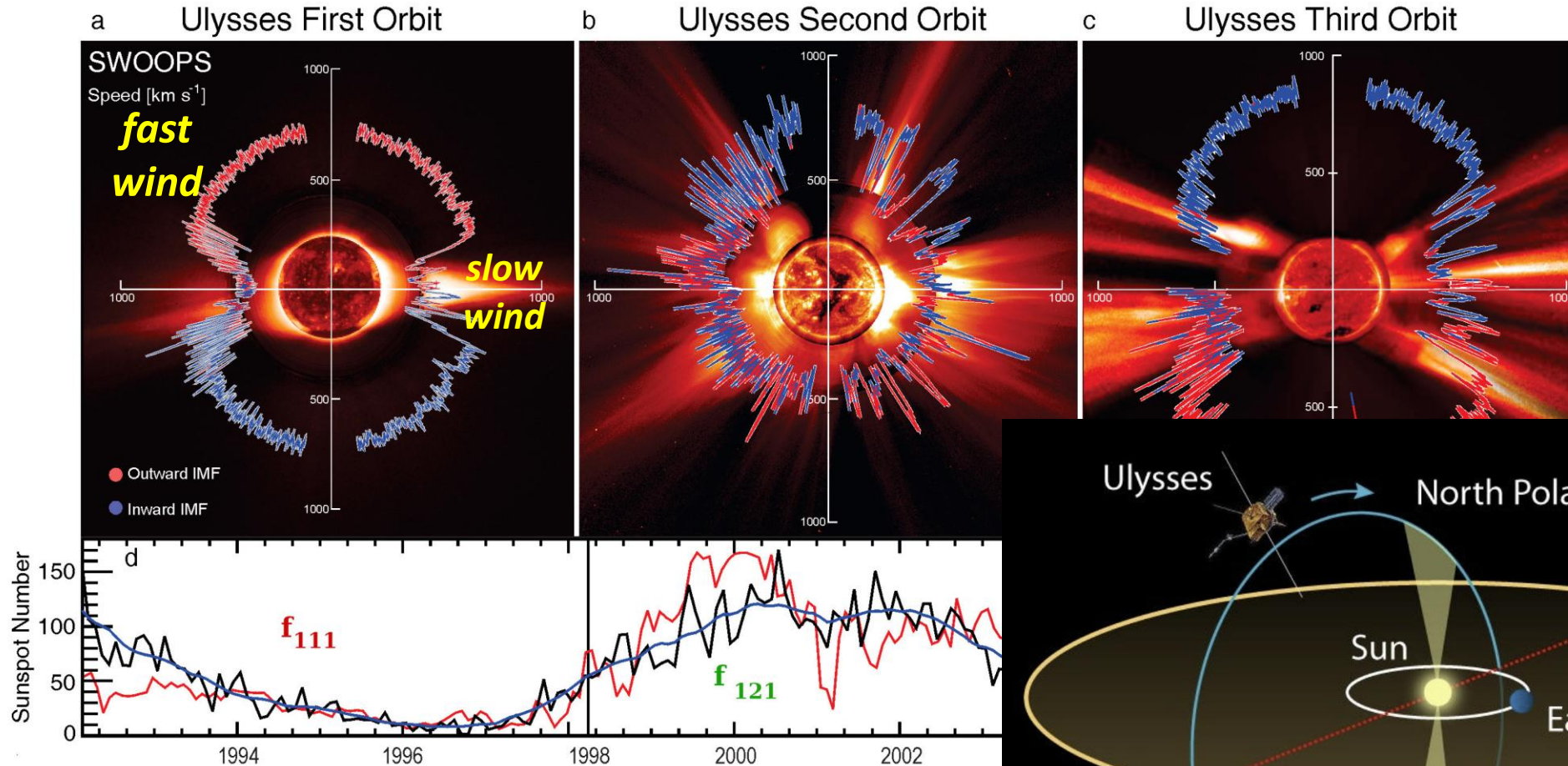
- ✓ 태양풍 근원
- ✓ 중이온 이온화 비율
- ✓ 연구의 의미 및 중요성

2. 태양풍 중이온 이온화 비율 예측 모델

- ✓ ACE 위성 데이터 1998 - 2011
- ✓ input paramters 선정 요인
- ✓ 중이온 이온화 비율 예측 모델 결과

3. 결론 및 논의

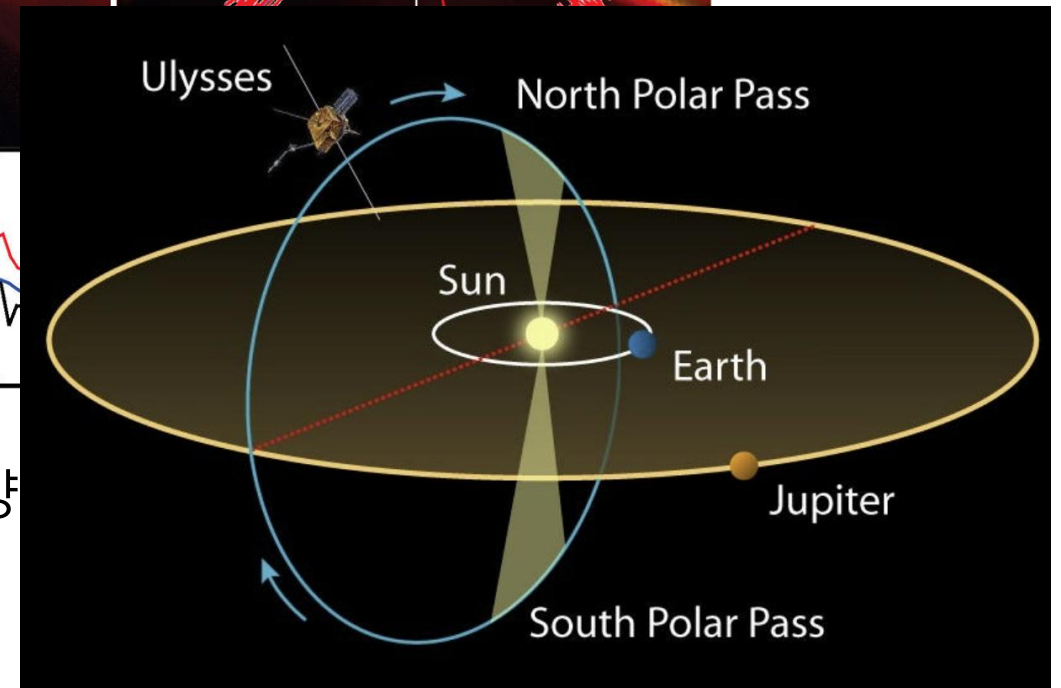
Solar wind by Ulyssess mission



흑점
개수

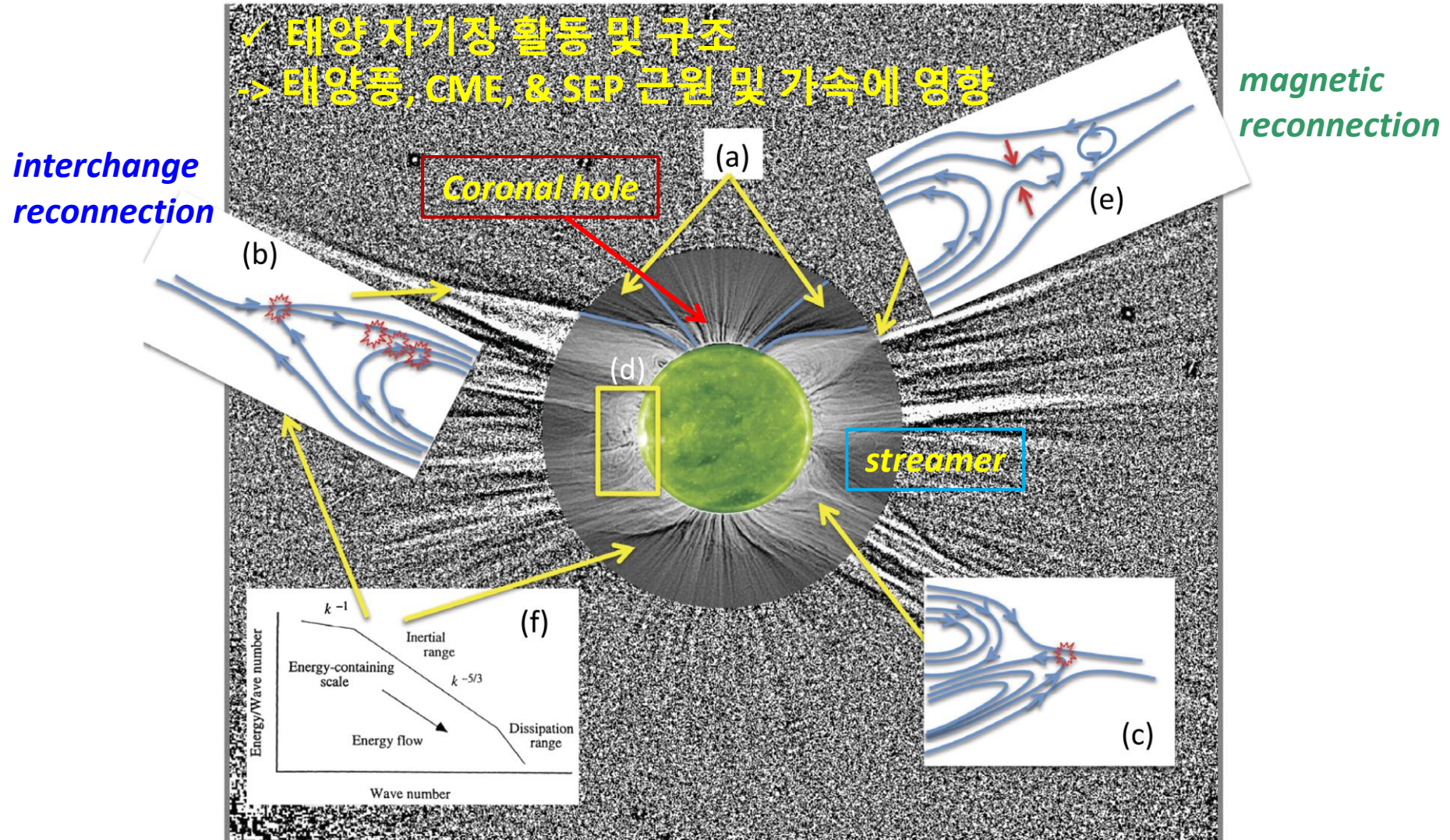
McComas et al., 2008, GRL

1990 ~ 2009년 태양



- ✓ **Solar wind structure**
depends mainly on the solar magnetic activity.

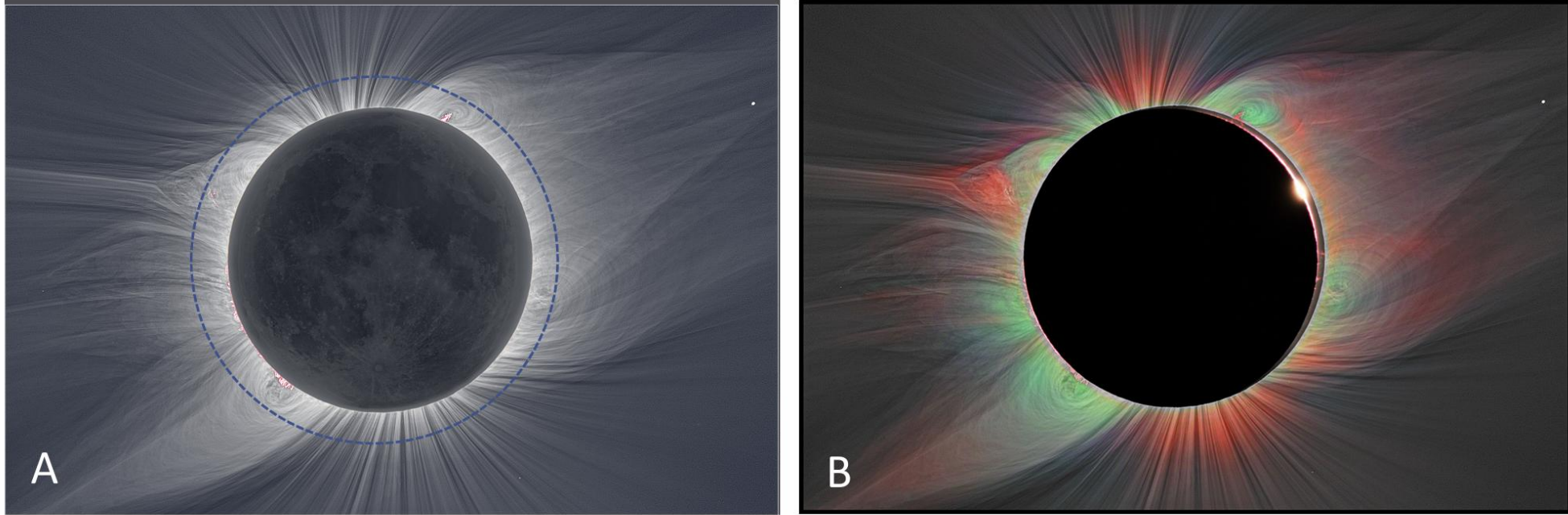
Possible sources of *slow* solar wind



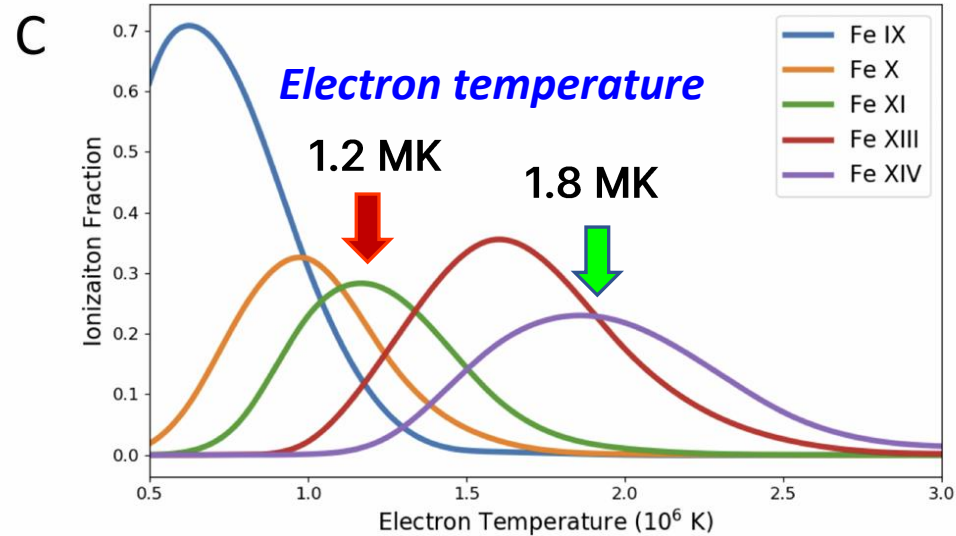
near the boundary layer
: *Alfven-wave turbulence*

Abbo et al., 2017, SP

Total Solar Eclipse 1 Aug 2008



Habbal et al., 2010, ApJ

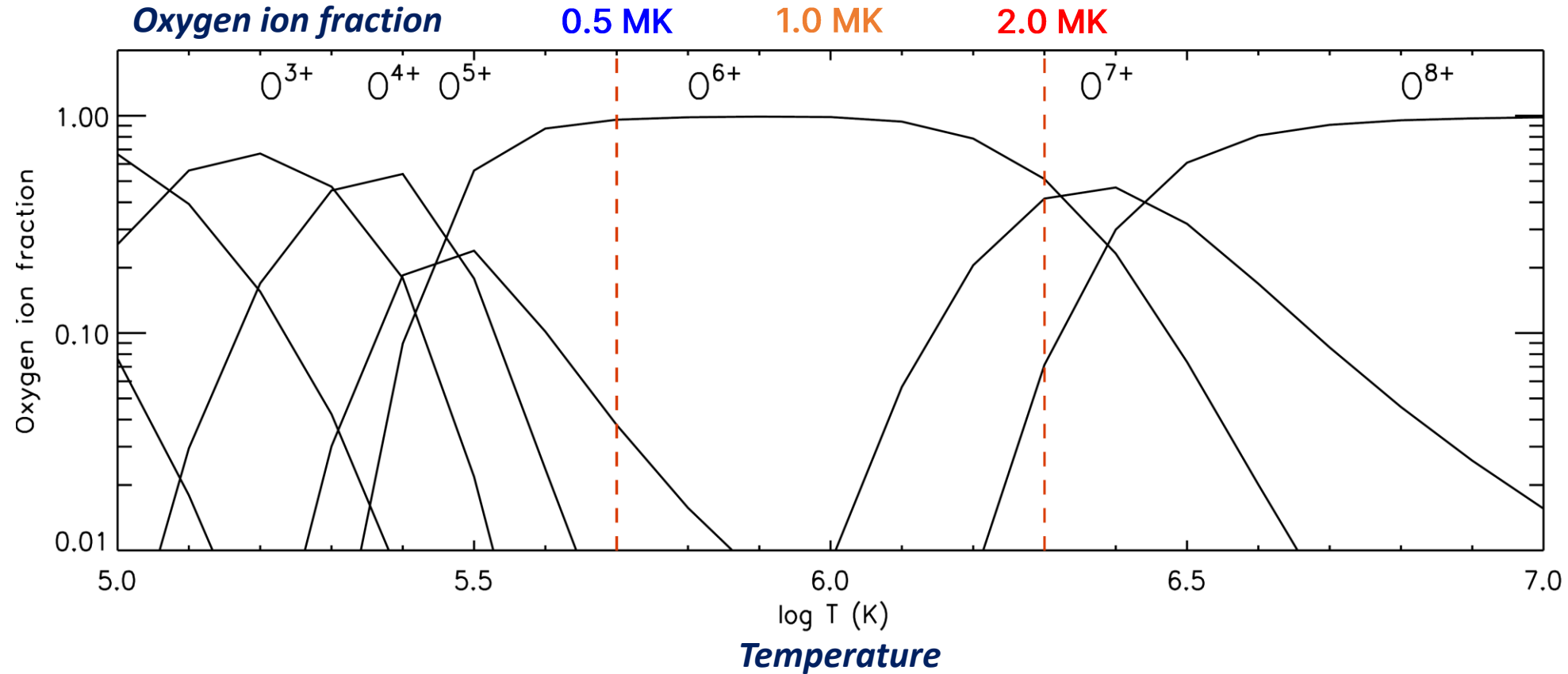
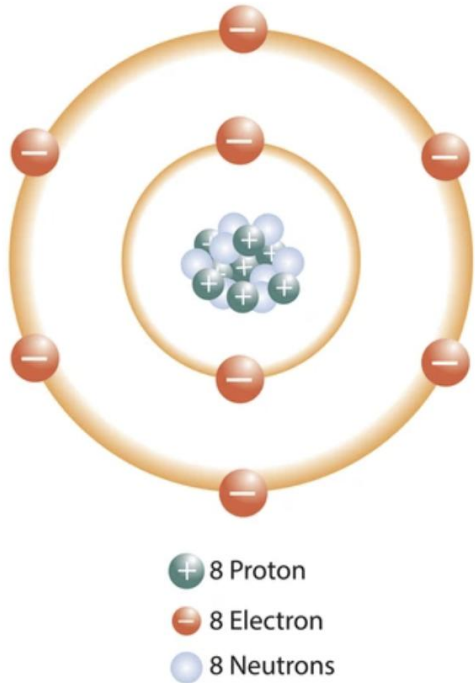


✓ Red: Fe X & Fe XI - **1.2 MK**

✓ Green: Fe XIII & Fe XIV - **1.8 MK**

Oxygen ion abundance as a function of temperature under ionization equilibrium

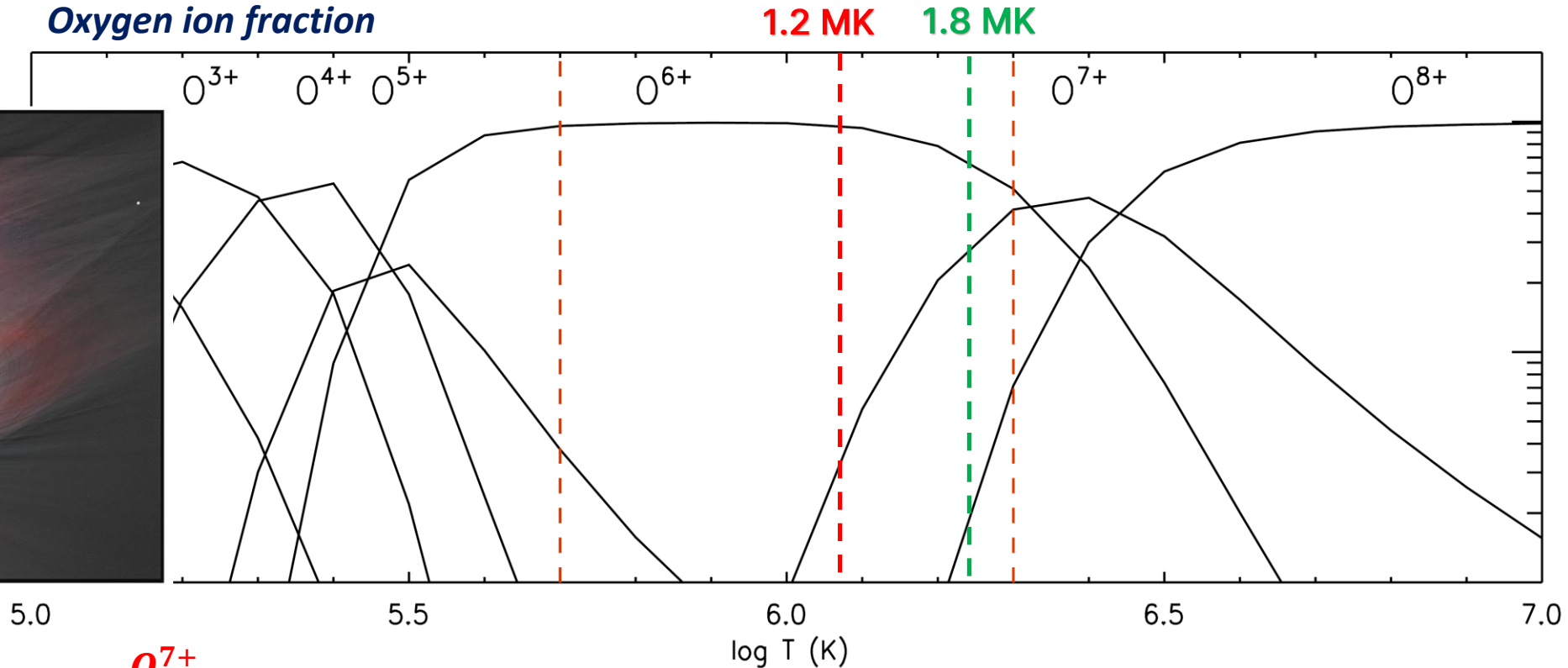
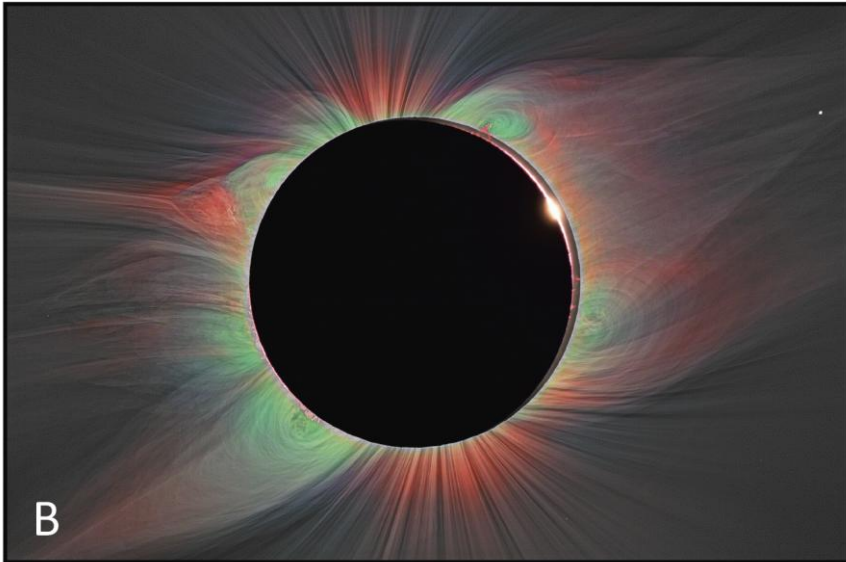
Oxygen Atom



- ✓ **Coronal temperature:** The Oxygen ionization state is "frozen-in" at a certain temperature as it expands from the corona.
- ✓ Higher coronal temperatures lead to higher charge states in the solar wind.

Oxygen ion abundance as a function of temperature under ionization equilibrium

Total Solar Eclipse 1 Aug 2008



✓ Red: Fe X & Fe XI - 1.2 MK

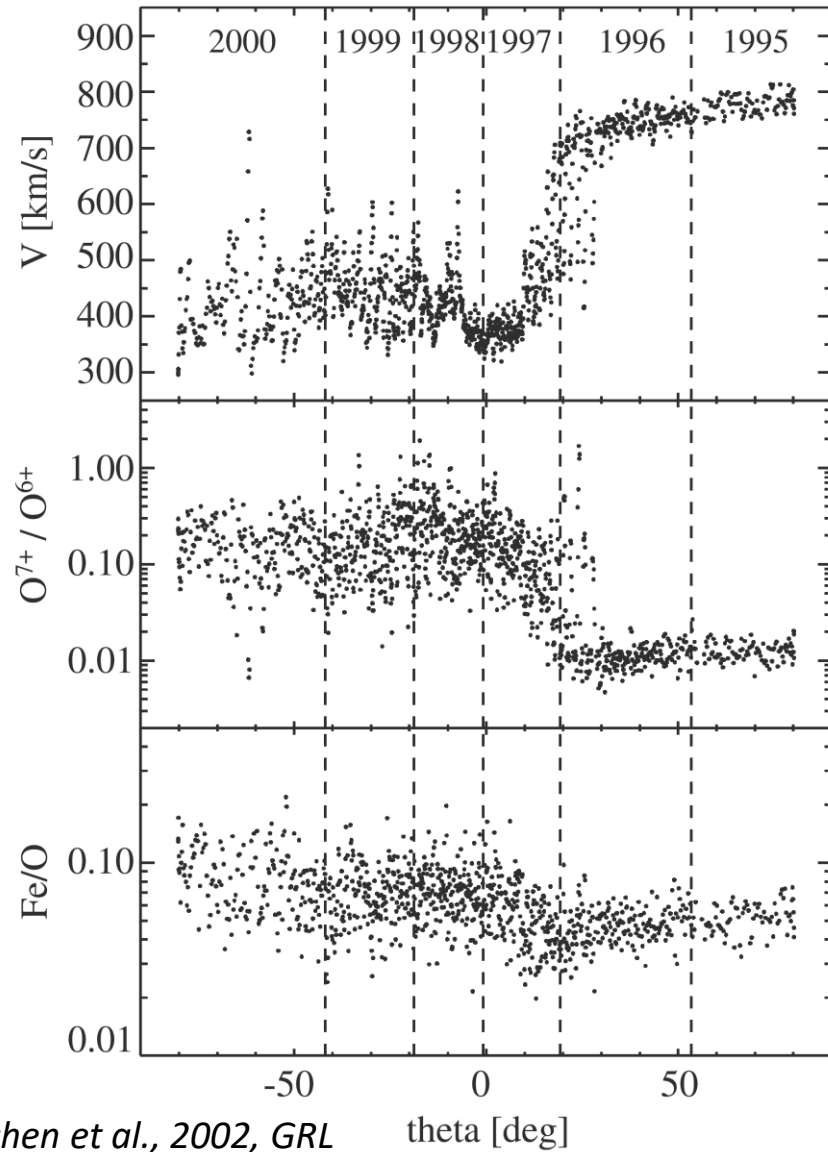
$$\frac{O^{7+}}{O^{6+}} \approx 0.05$$

✓ Green: Fe XIII & Fe XIV - 1.8 MK

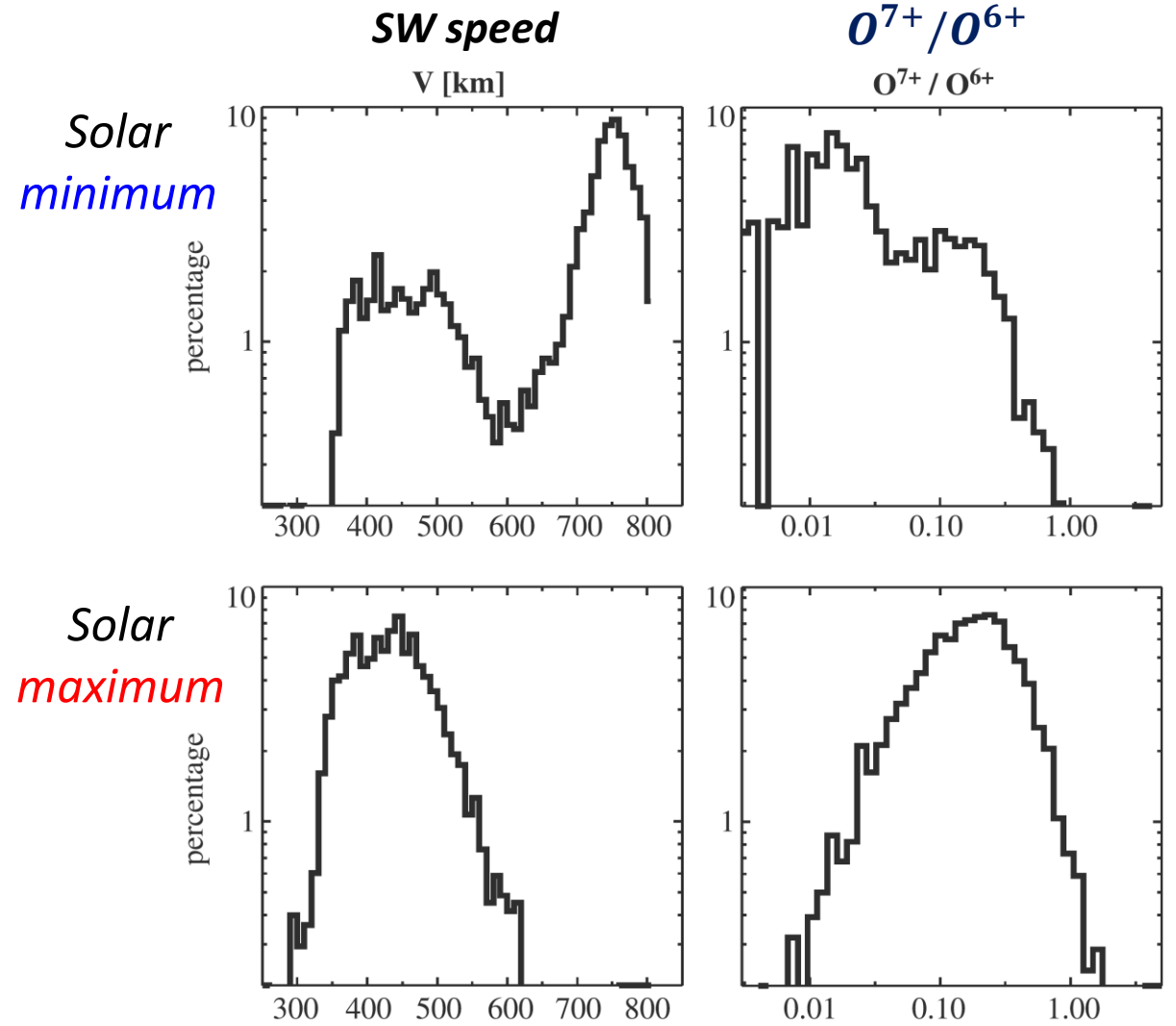
$$\frac{O^{7+}}{O^{6+}} \approx 0.5$$

✓ It is expected that O^{7+}/O^{6+} are inversely correlated with wind speed.

Ulysses observation



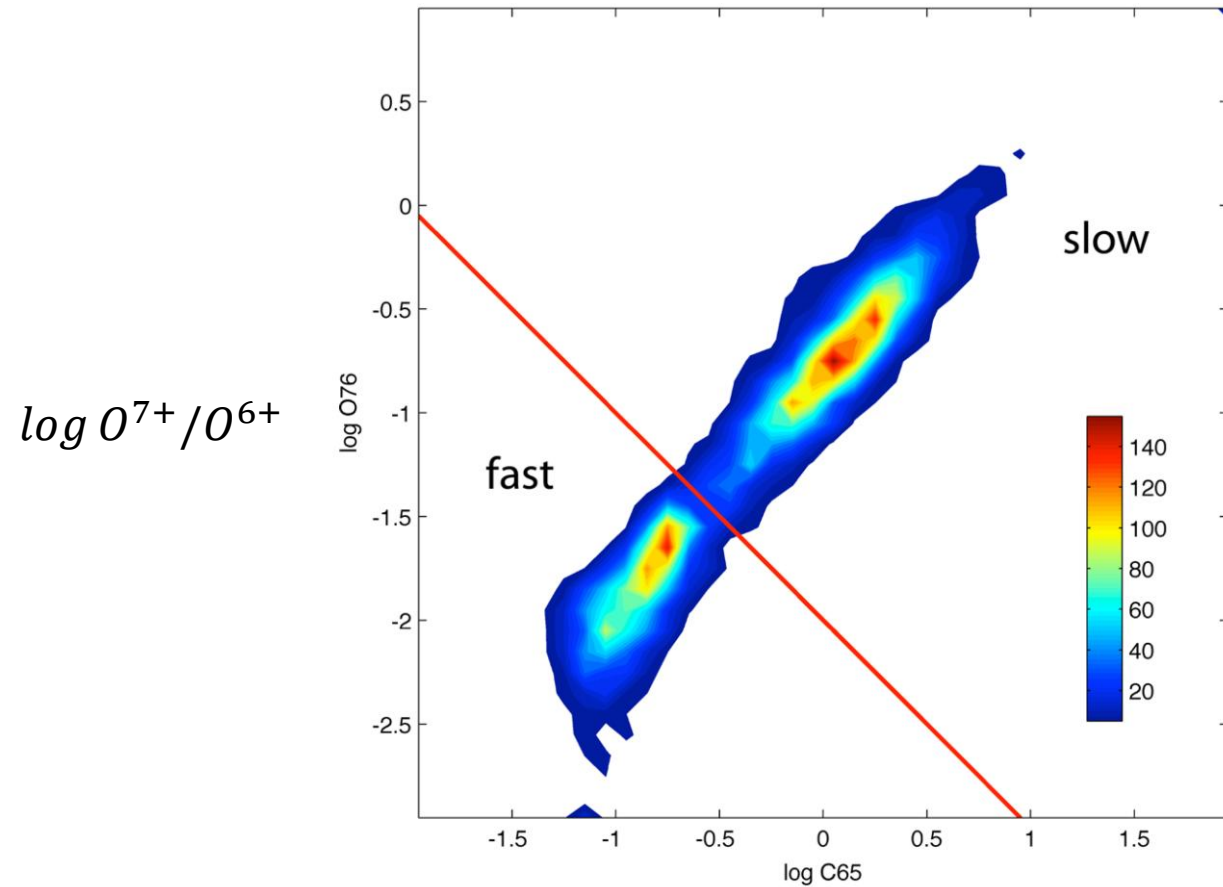
Zurbuchen et al., 2002, GRL



✓ **Fast solar wind**, originating from coronal holes, has consistently lower charge states.

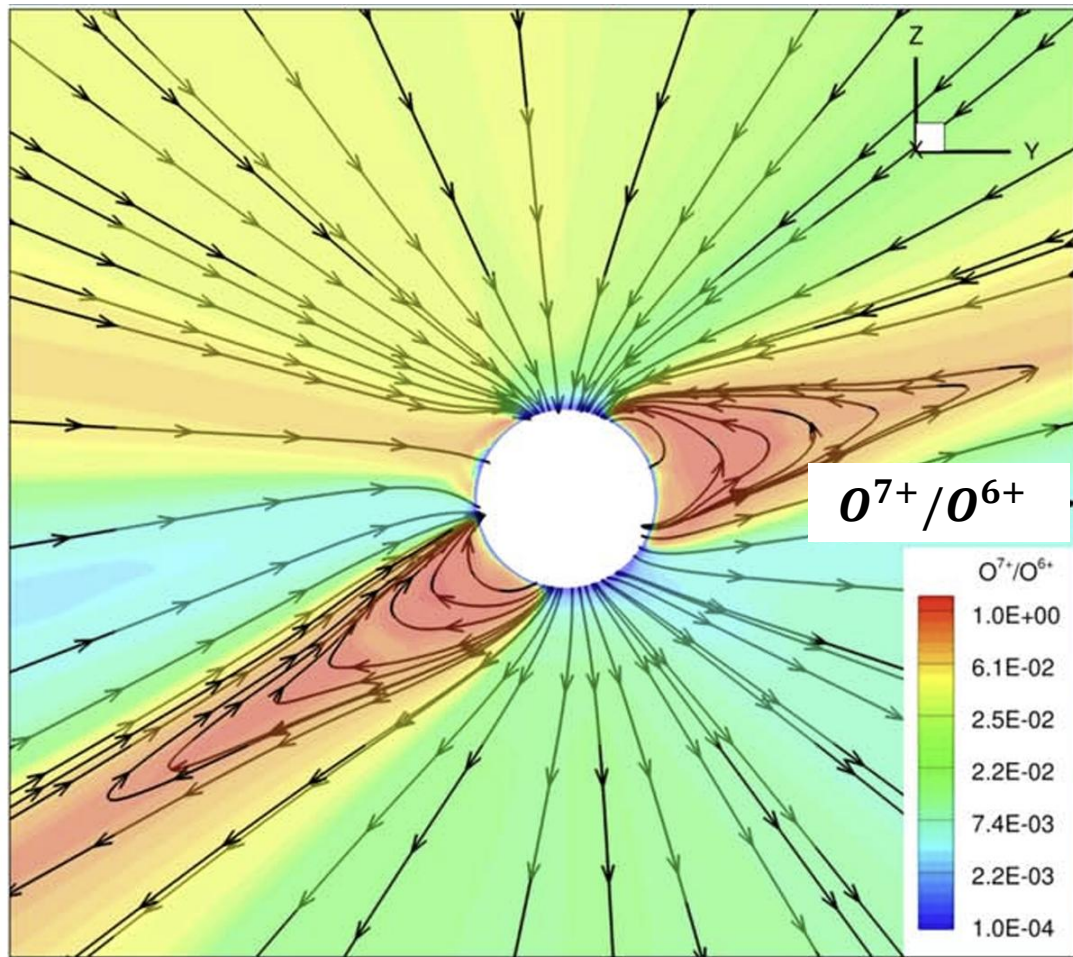
Ulysses observation

Oxygen ratio (O^{7+}/O^{6+}) and Carbon ratio (C^{6+}/C^{5+})

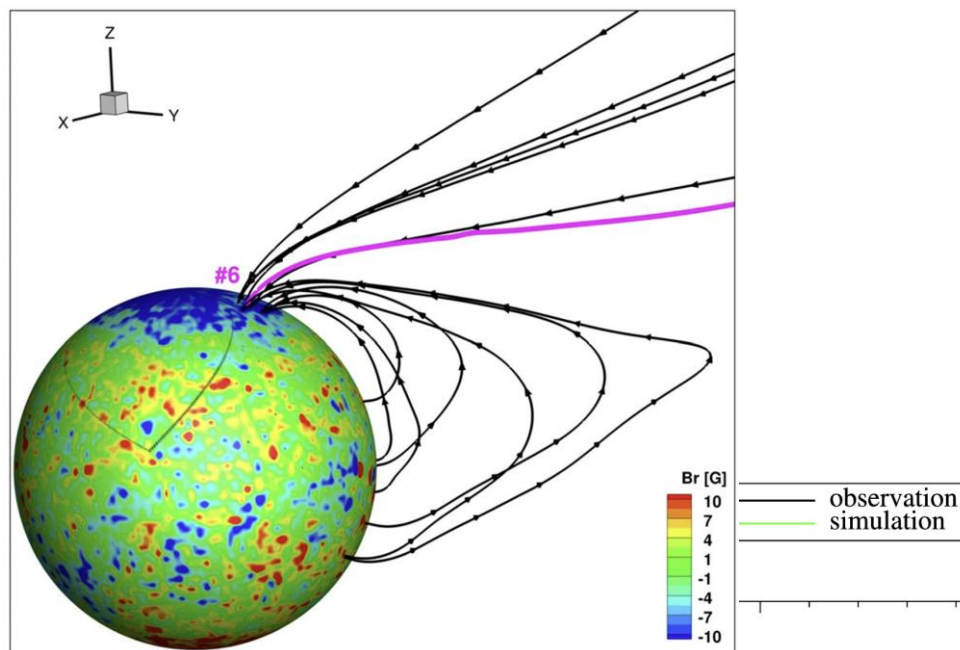


Abbo et al., 2016, SSR

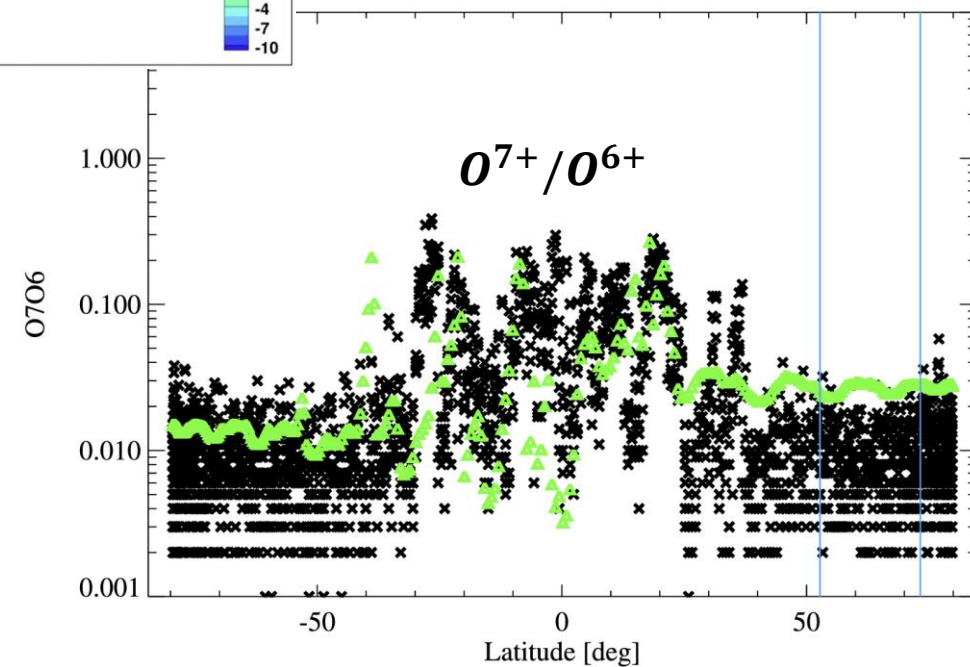
연구 의미 및 중요성 1 : 태양풍 근원



Charge state calculation
for global solar wind modeling

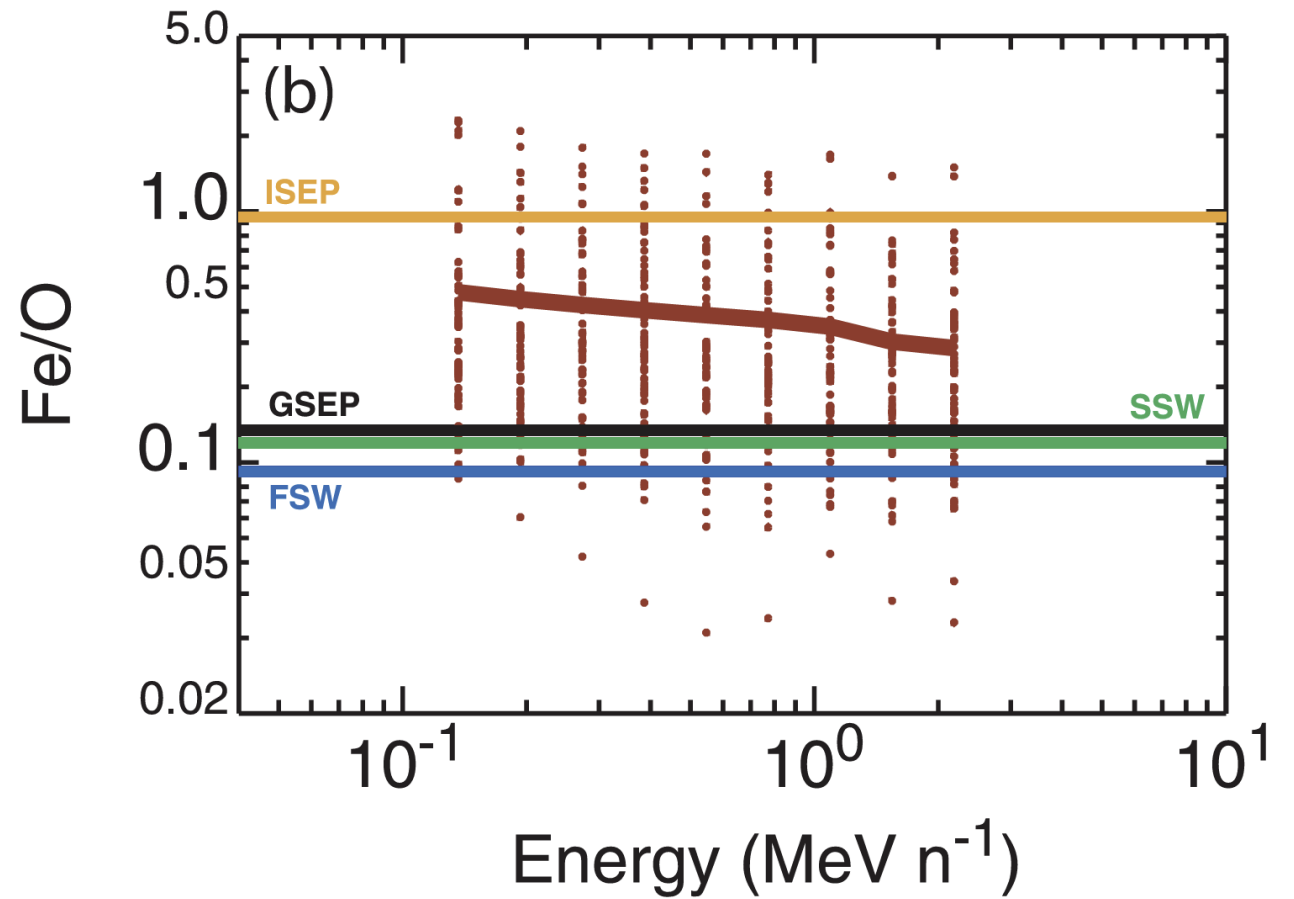
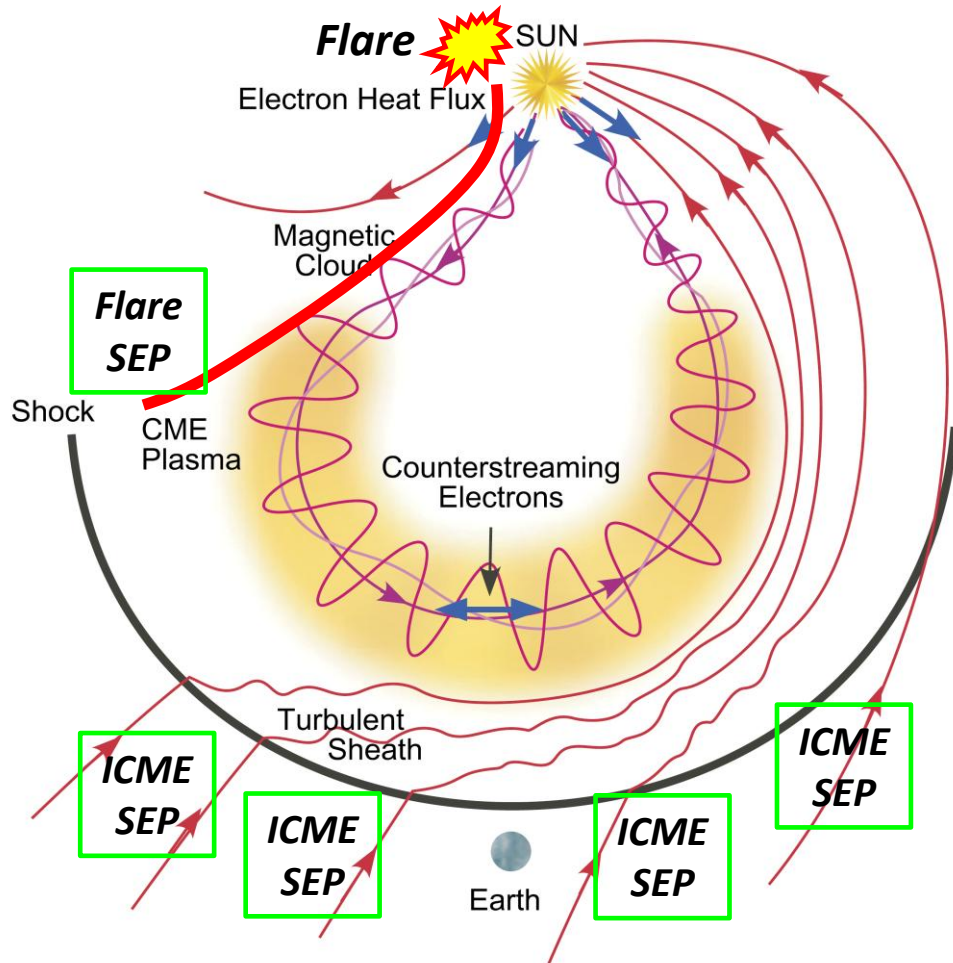


Szente et al., 2022, ApJ



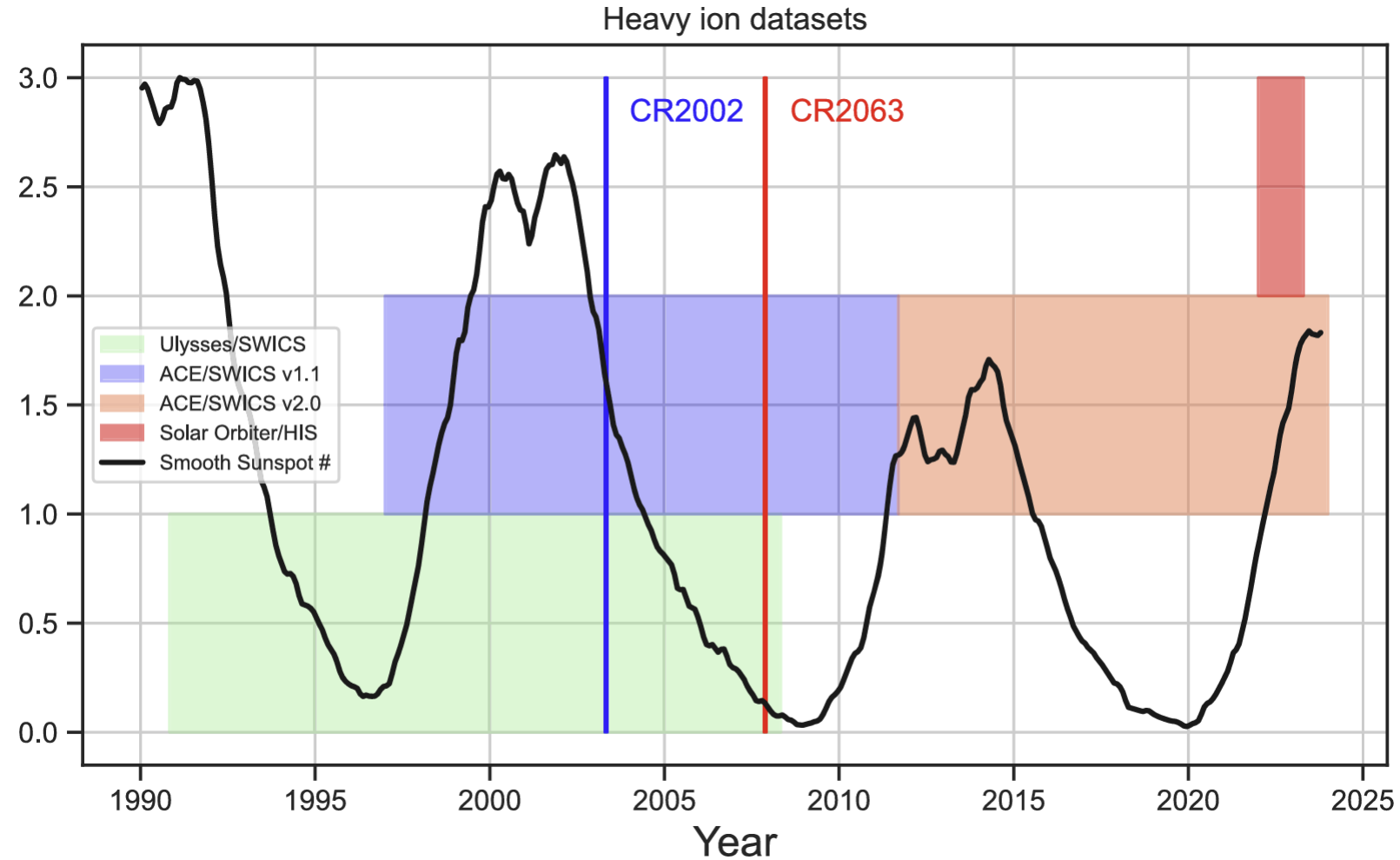
연구 의미 및 중요성 1 : 태양풍 근원

Standard picture of "ICME"



연구 의미 및 중요성 2: 중이온 현장 관측기

*Timeline exhibiting for availability of **heavy ion in situ measurements***



Data set for deep learning

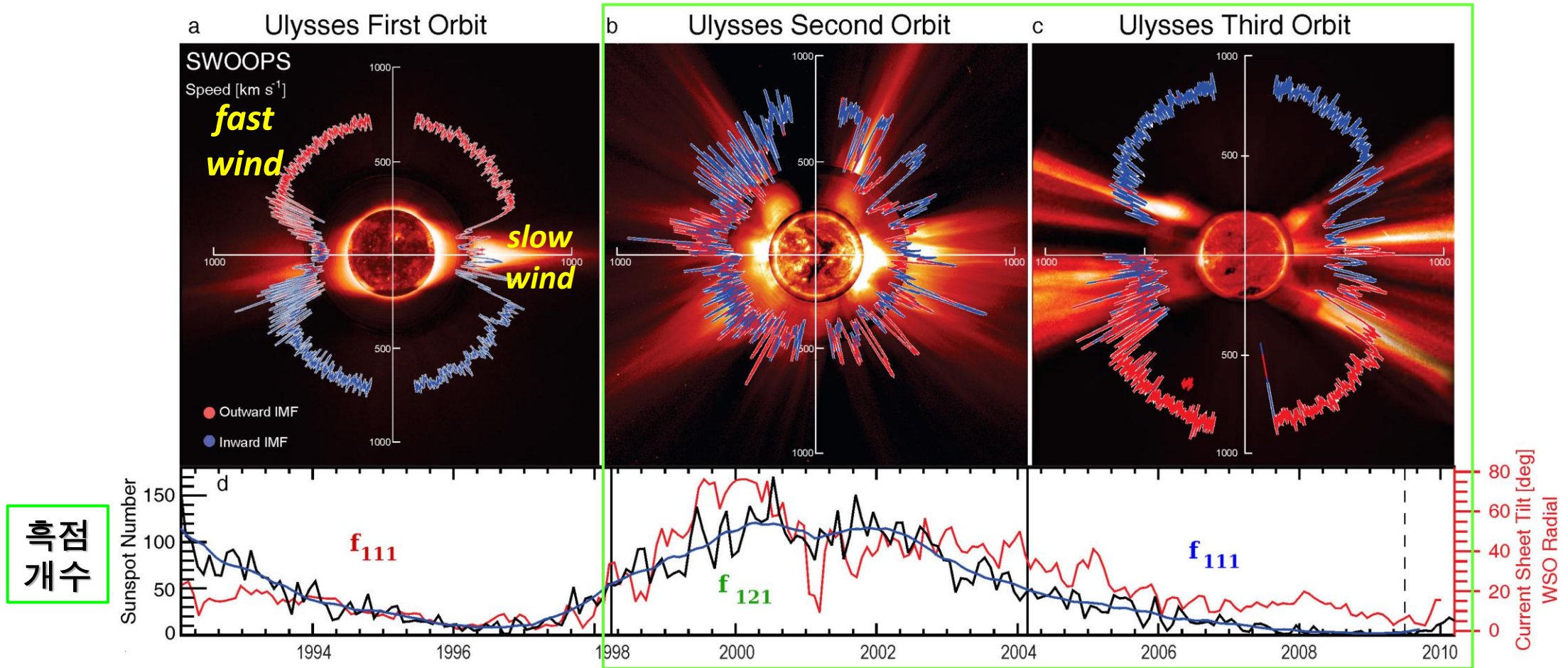
1. ACE spacecraft : 태양풍 양성자 및 자기장 자료

- ✓ **Input** data (64초): Solar wind speed, density, temperature, helium abundance, proton specific entropy, Coulomb number, & Alfvenicity (SWEPAM & MAG)
- ✓ **Target** data (2시간): O^{7+}/O^{6+} , C^{6+}/C^{5+} , & Fe/O - 중이온 이온화 비율 (SWICS)

2. Period: Feb 1998 – Aug 2011, 태양 극대/극소기 포함 약 13년

- ✓ Training set: 매년 1 – 8월
- ✓ Validation set: 매년 9월
- ✓ Test set: 매년 10 – 12월
- ✓ Target data 개수: 12개(1일) x 365일(1년) x 13년 ~ 약 50,000개

Data set for deep learning

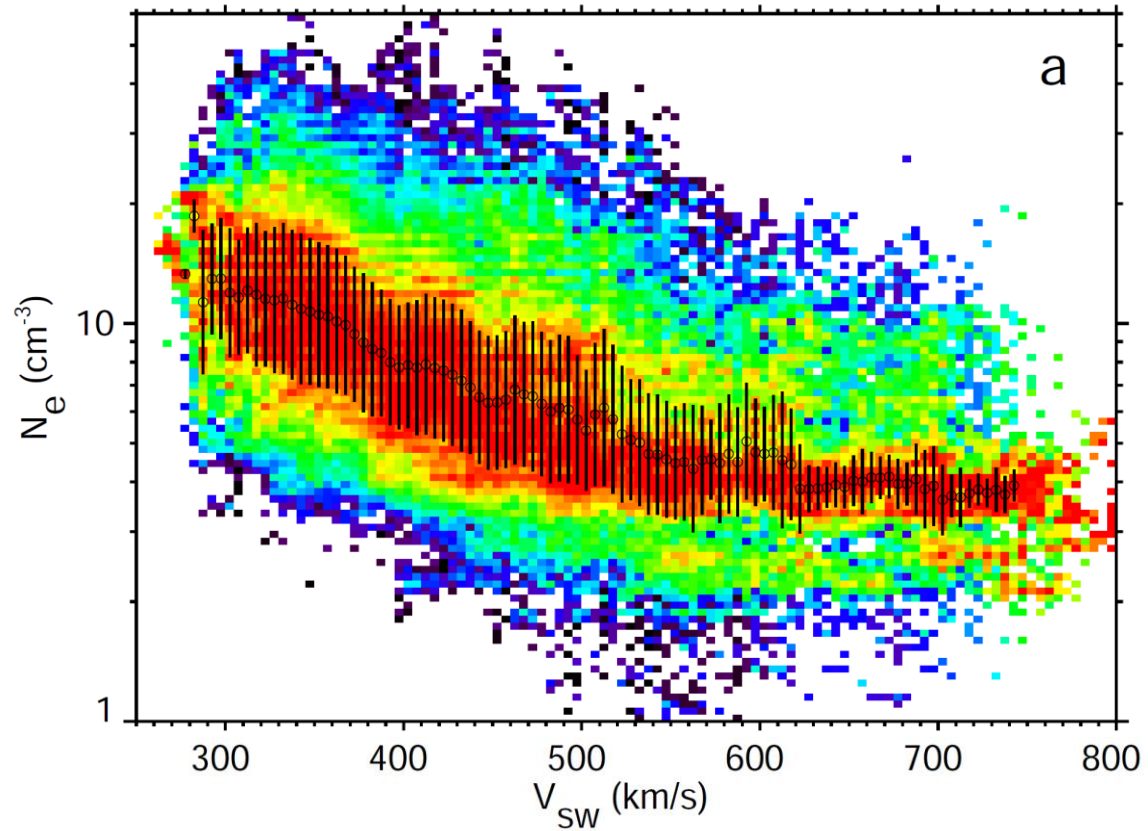


McComas et al., 2008, GRL

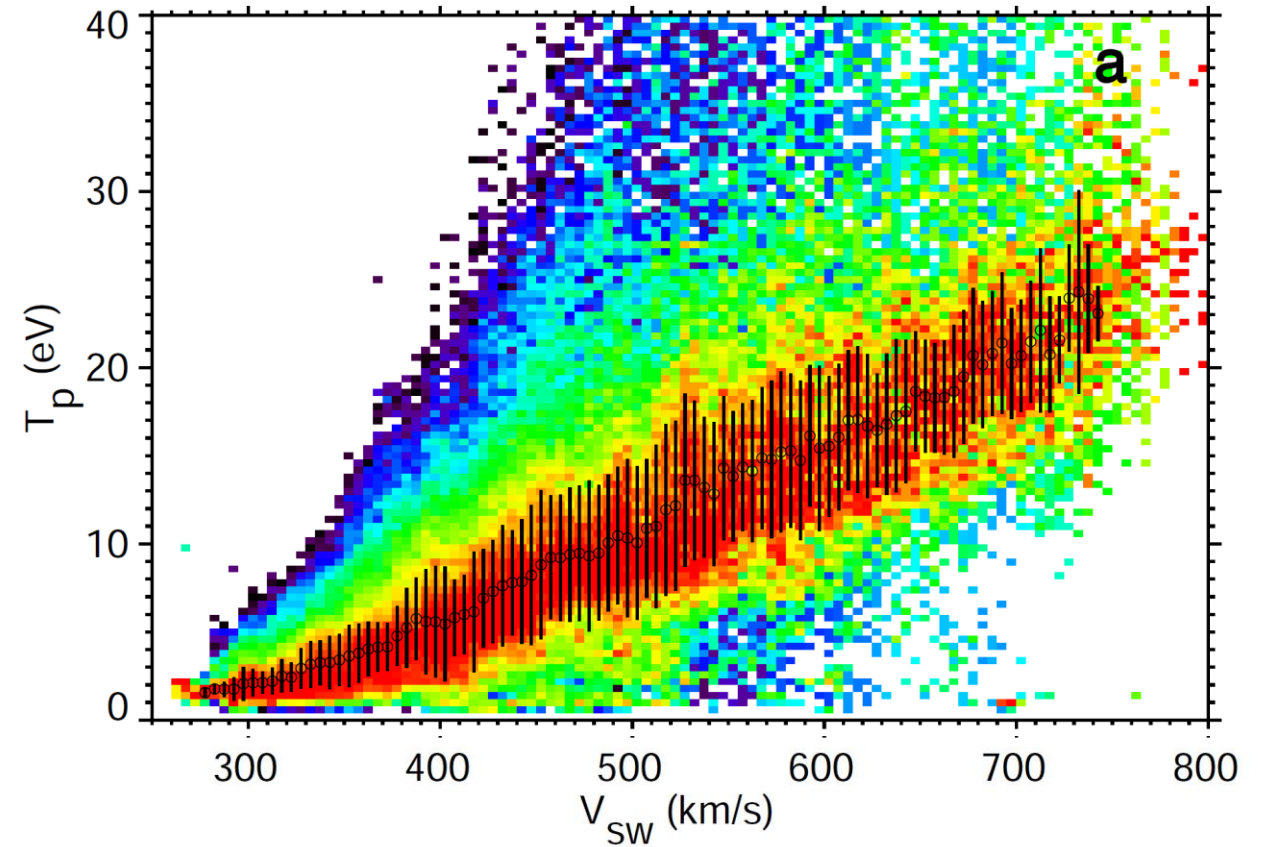
1998 ~ 2011년 태양 극대/극소기 포함 약 13년

Statistic results for input data @ 1 AU

Speed vs Density



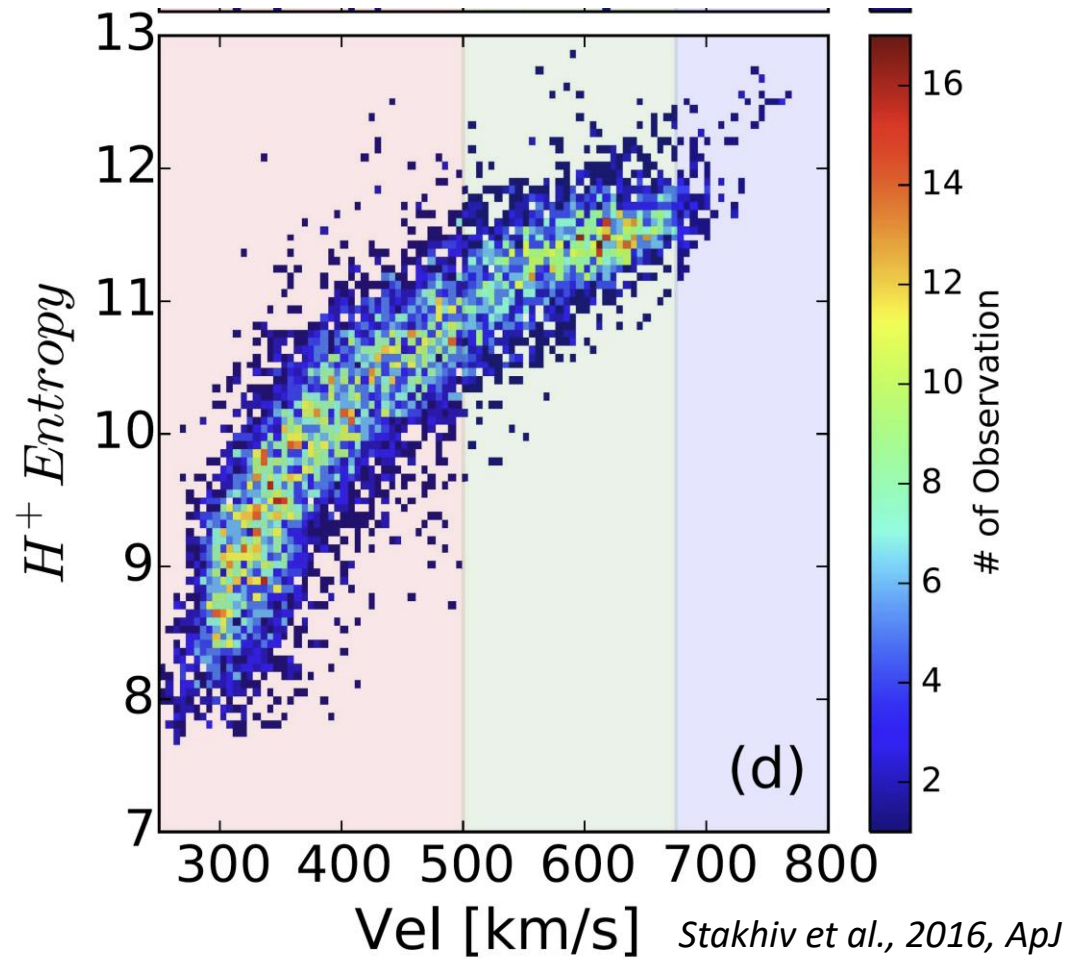
Speed vs Temperature



Salem et al., 2023, A & A

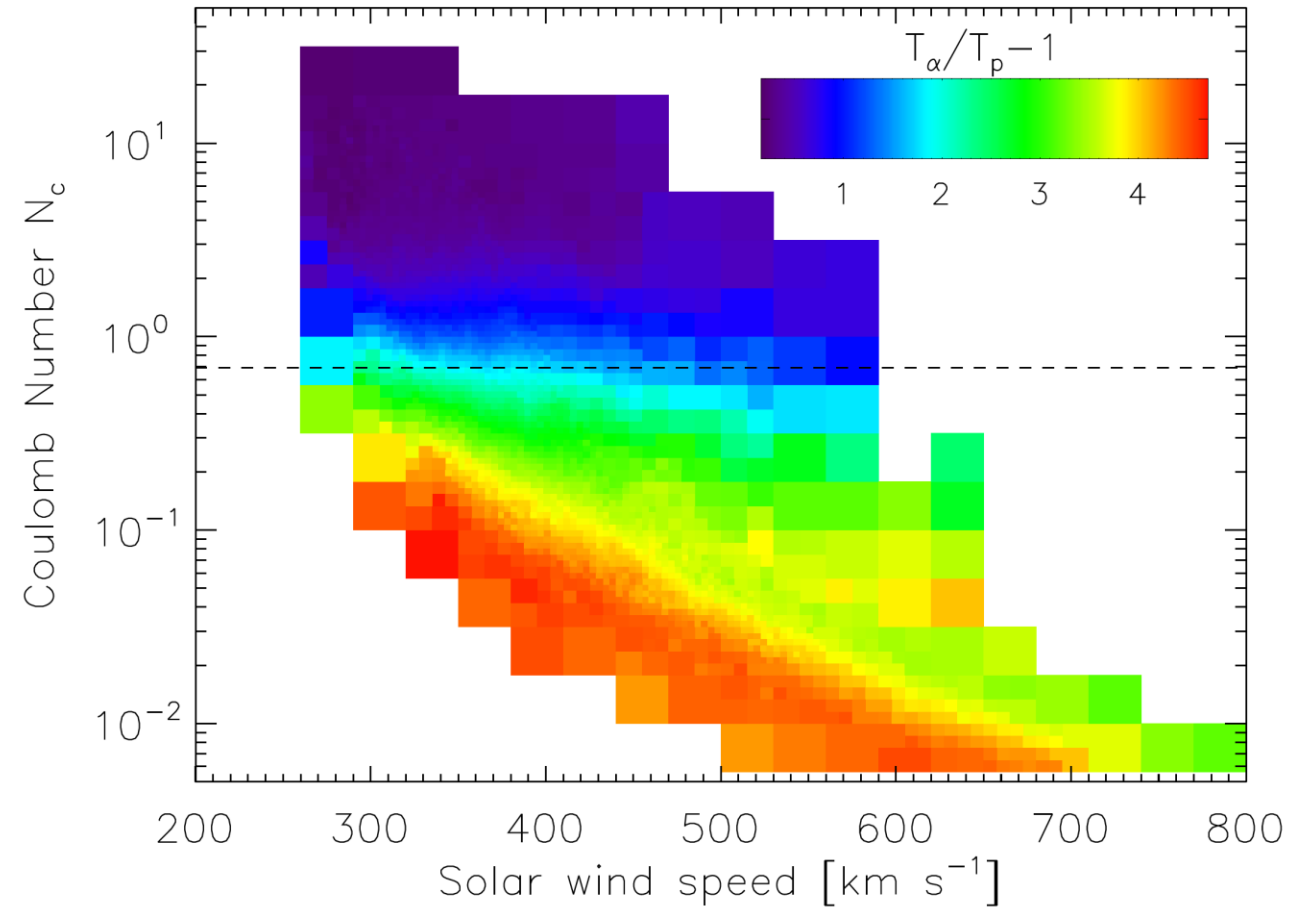
Statistic results for input data @ 1 AU

Speed vs Entropy



✓ Proton specific entropy: $\ln\left(\frac{T_p}{n^{1/2}}\right)$

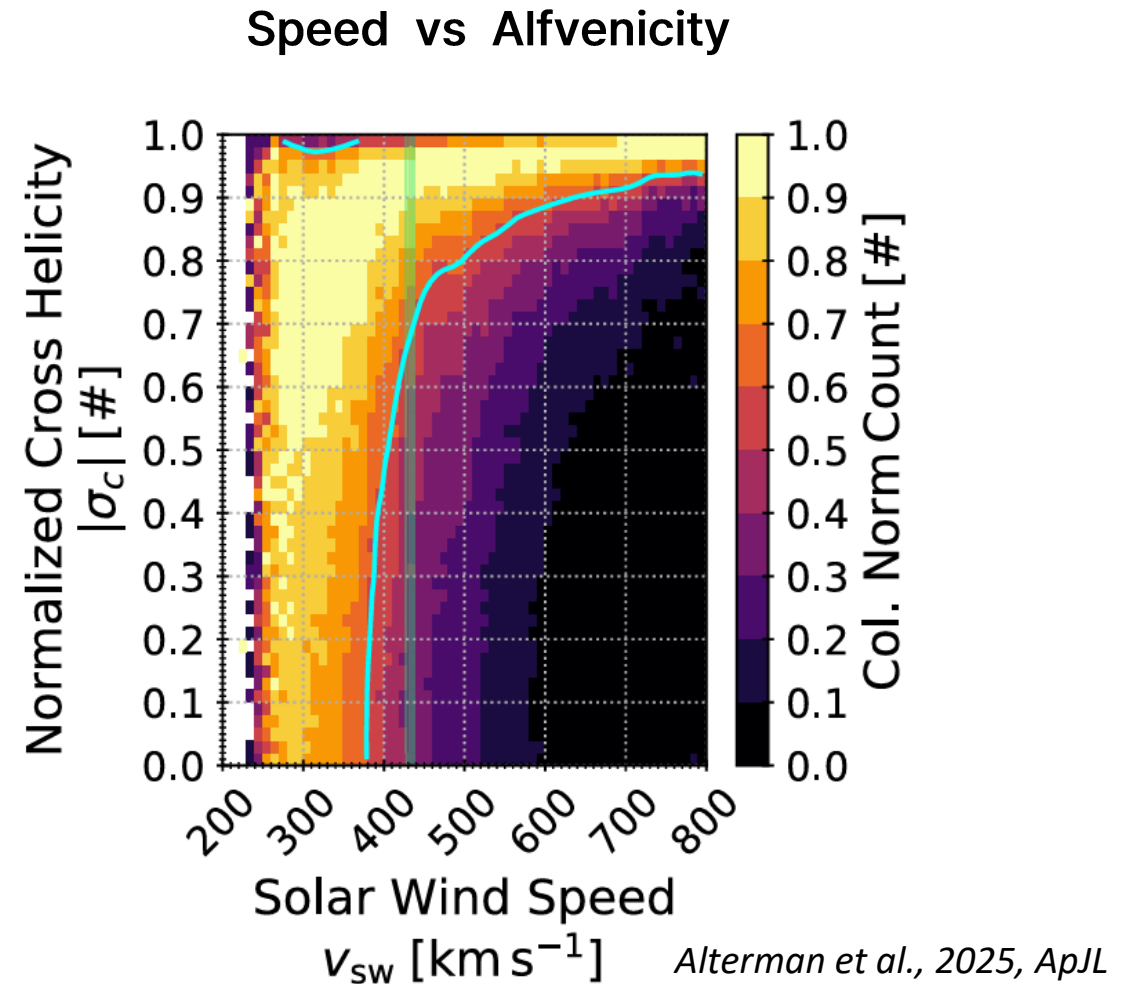
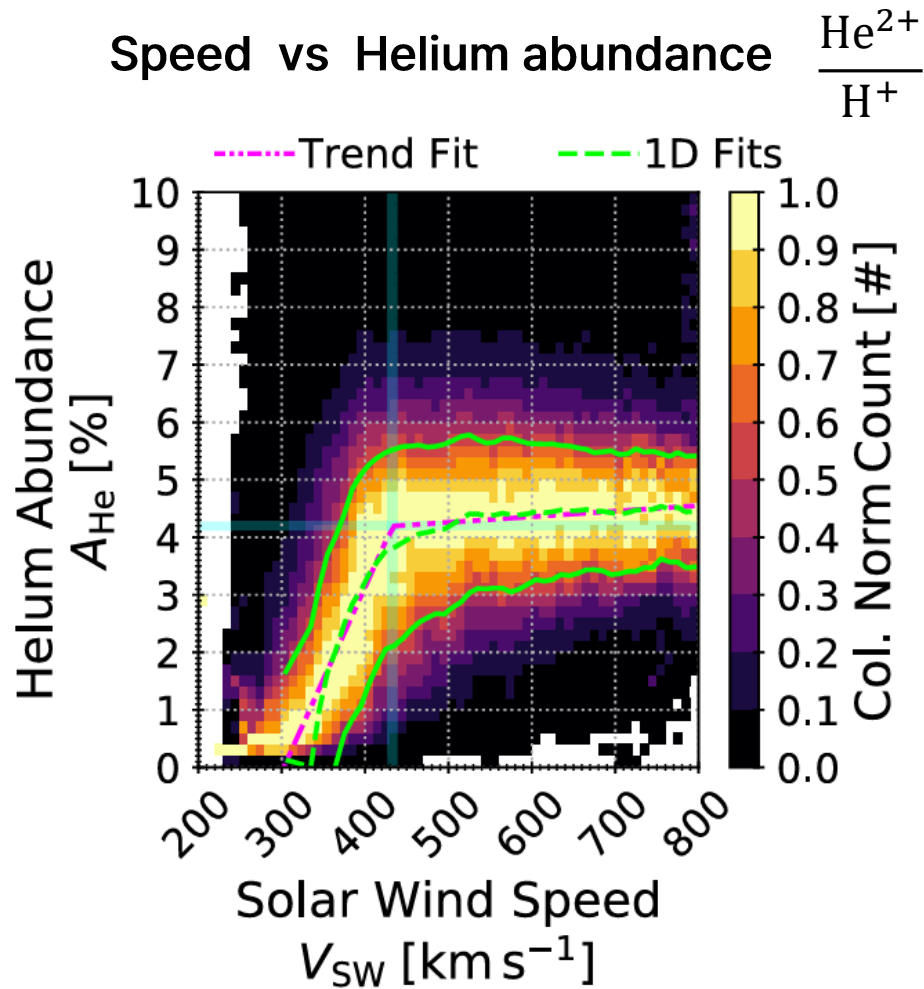
Speed vs Coulomb collisions



Kasper et al., 2017, *ApJ*

✓ Collisional frequency: $\frac{n_p}{T_p^{3/2}}$

Statistic results for input data @ 1 AU



Alfvenicity

- *Alfvenicity is a characteristic property of the observed magnetic fluctuations, namely a high-degree of magnetic field (b)-velocity (v) correlations,*

$$-1 < \text{Alfvénicity} < 1$$

$$\sigma_c = 2 \langle \delta \mathbf{V} \cdot \delta \mathbf{V}_A \rangle / \langle \delta \mathbf{V}^2 + \delta \mathbf{V}_A^2 \rangle$$

$$(\text{i.e., } \delta \mathbf{V}_A = (\mathbf{B} - \langle \mathbf{B} \rangle) / \sqrt{\mu_0 \langle \rho \rangle}).$$

$$\delta \mathbf{V} = \mathbf{V} - \langle \mathbf{V} \rangle$$

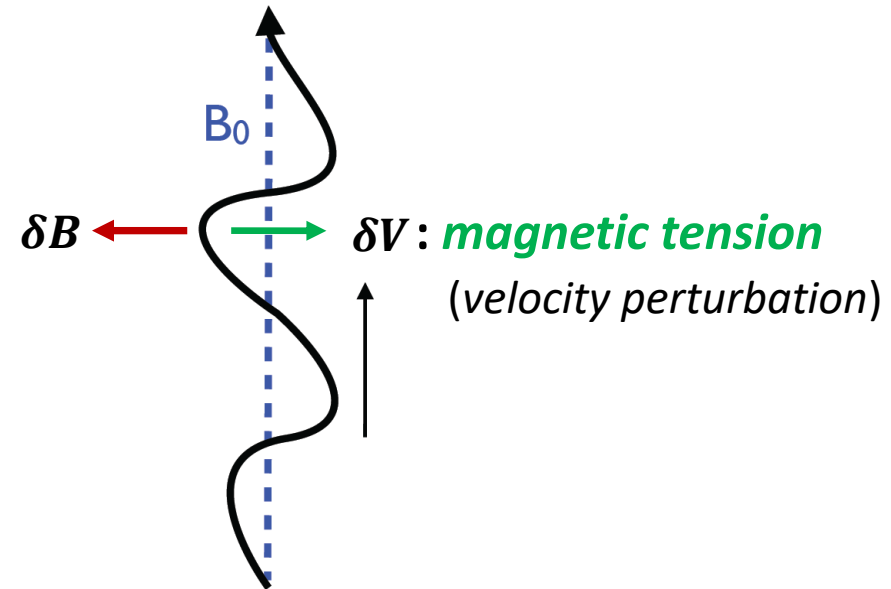
as in large amplitude Alfvén waves
“*weak compressibility*”

$$\delta n_p / n_p \ll (\delta v / V_{th})^2,$$

$$V_A = B_0 / (4\pi m_p n_p)^{1/2} : \text{Alfvén velocity}$$

$$V_{th} = (2k_B T / m_p)^{1/2} : \text{thermal velocity}$$

$$|\mathbf{B}| = \text{constant}$$

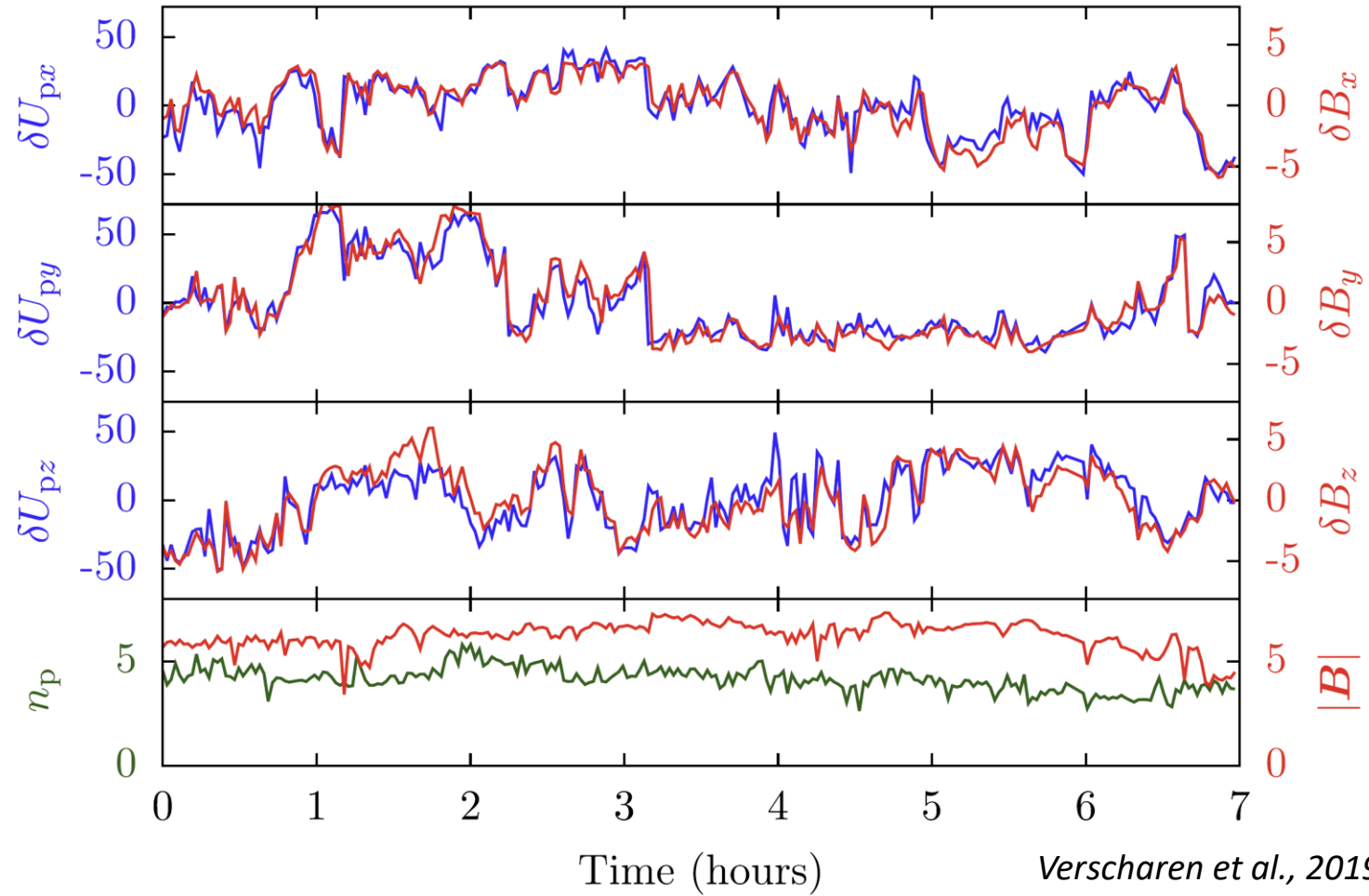


Alfvén waves

Alfvénic correlations
for *incompressive turbulence*

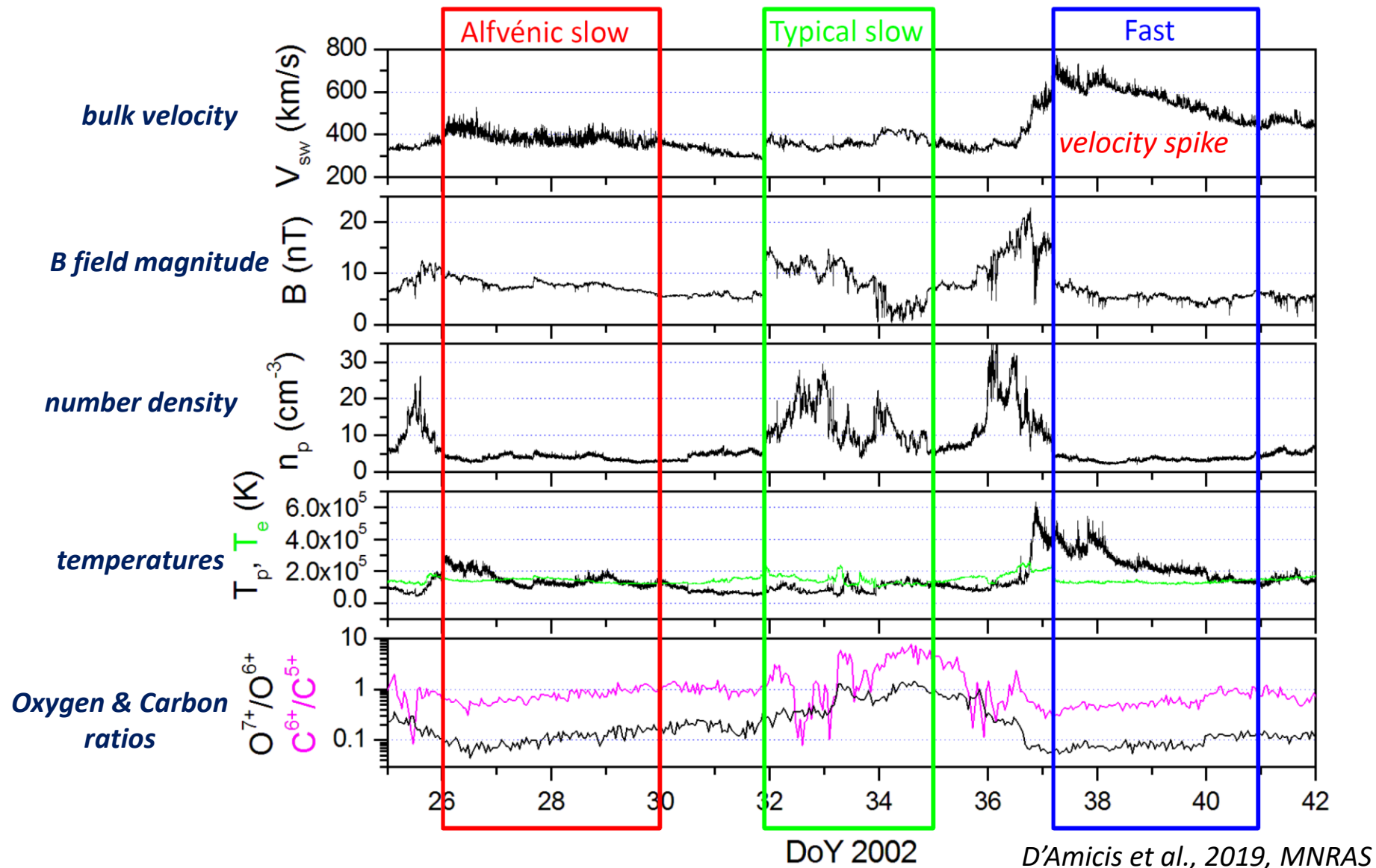
$$\delta \mathbf{v} \simeq \pm \frac{\delta \mathbf{B}}{\sqrt{4\pi\rho}} \quad \text{sign}[-\mathbf{k} \cdot \mathbf{B}_0]$$

Alfvenic correlation in the solar wind



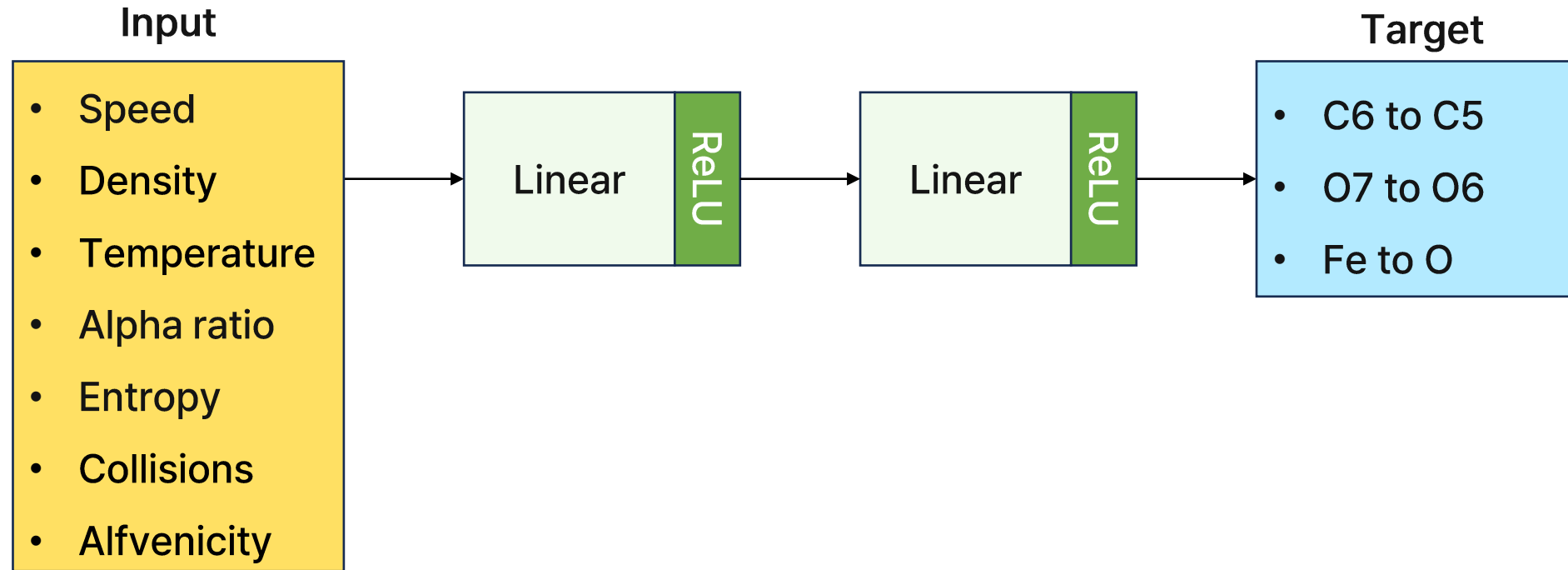
Typical example of three types solar wind

WIND spacecraft measurement at 1 AU



Deep learning model description

Model Simple MLP



Loss function weighted MSE

$$\text{weight} = 1 + (\alpha * \text{target})$$
$$\text{loss} = \text{weight} (\text{target} - \text{prediction})^2$$

In this experiment, we set α to 3.

Optimizer Adam with learning rate 0.0005

Experiment	Correlation Coefficient			Input parameters
	C^{6+}/C^{5+}	O^{7+}/O^{6+}	Fe/O	
Ex1	0.43	0.67	0.54	Alfvenicity 절대값 없이
Ex2	0.51	0.65	0.52	Vp, Alpha_ratio, entropy, Alfvenicity , Np, Tp , coulomb num
Ex3	0.25	0.67	0.55	Vp, Alpha_ratio, entropy
Ex4	0.33	0.69	0.55	Alfvenicity 제외
Ex5	0.50	0.59	0.44	Vp 제외
Ex6	0.41	0.67	0.54	Alpha ratio 제외
Ex7	0.45	0.62	0.49	Entropy 제외
Ex8	0.38	0.64	0.53	Np 제외
Ex9	0.33	0.64	0.55	Tp 제외
Ex10	0.38	0.66	0.50	Coulomb_num 제외
Ex11	0.41	0.59	0.46	Vp, Alpha_ratio, Alfvenicity , Np, Tp

Table 1. Results of solar wind ion prediction experiments with varying input parameters.

- ✓ 전반적으로 O^{7+}/O^{6+} 결과의 성능이 좋음.
- ✓ 물리량 계산값을 추가한 것이 모델 학습 결과가 더 좋음. 22

Deep learning results

Experiment 2 - Vp, Alpha_ratio, entropy, |Alfvenicity|, Np, Tp, coulomb num

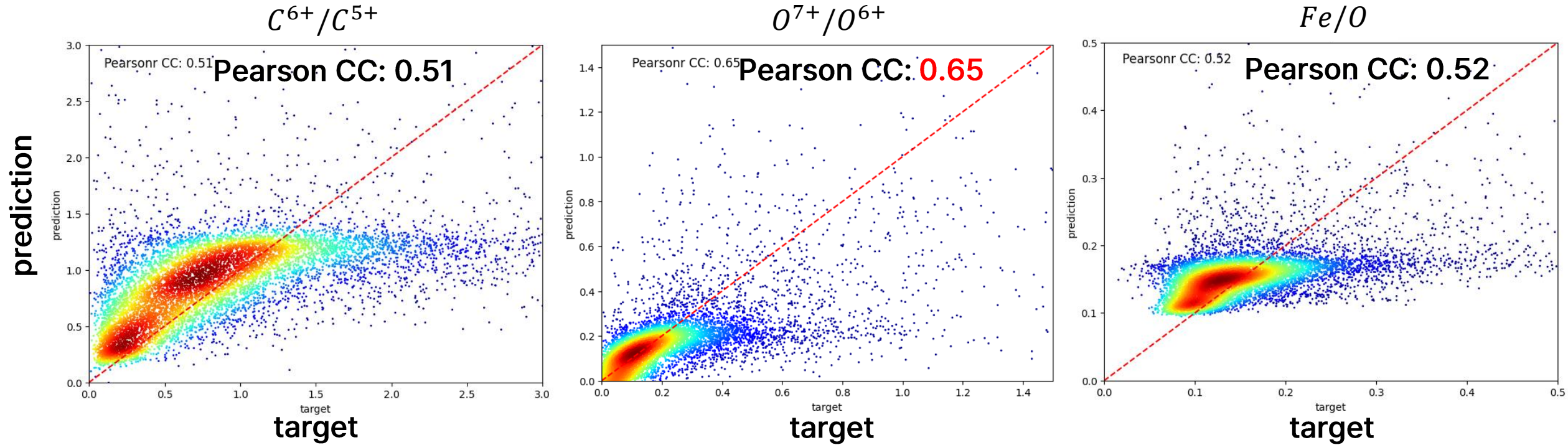


Figure 1. Scatter density plots of predicted [from Experiment 2 in Table 1](#) versus observed ion composition ratios for three species: C^{6+}/C^{5+} (left), O^{7+}/O^{6+} (middle), and Fe/O (right). The red dashed line represents one-to-one line, and the Pearson correlation coefficient (CC) is shown in each panel.

ElasticNet: linear regression

일반적인 선형 모형에 가중치 w 를 구할 때 목적 함수에 규제항(regularization) 추가

$$\hat{y} = b + \sum_{j=1}^d x_j w_j$$

b: 절편
 x_j : 입력 feature 개수 (우리의 경우에는 시간 7 x 변수 7 = 49개)
 w_j : 각 feature에 곱해지는 회귀 계수

Experiment				Input parameters
	C^6/C^5	O^7/O^6	Fe/O	
Linear Regression	0.36	0.59	0.41	Vp, Alpha_ratio, entropy, Alfvénicity , Np, Tp , coulomb num

Table 2. Results of solar wind ion prediction using linear regression with same input parameters as Ex2 in Table 1.

- ✓ Linear regression 결과는 딥러닝 결과보다 더 낮은 CC 를 보여줌.
- ✓ 딥러닝 모델이 linear regression이 포착하지 못한 non-linearity를 학습했을 것.

Linear regression results

✓ Deep learning
results

Pearson CC: 0.51

Pearson CC: 0.65

Pearson CC: 0.52

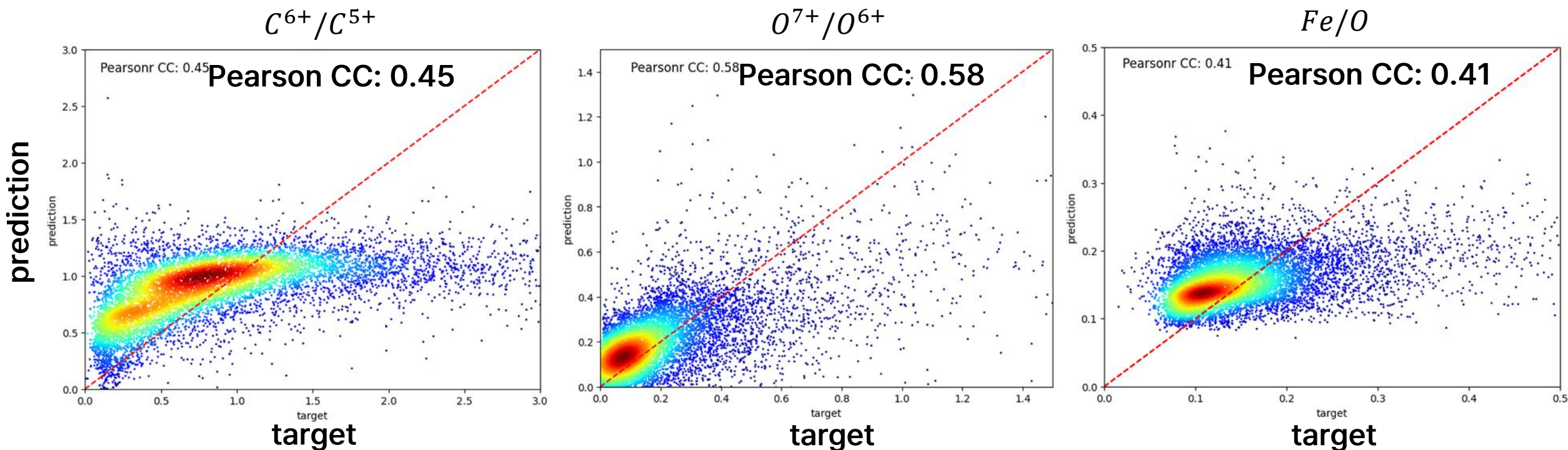
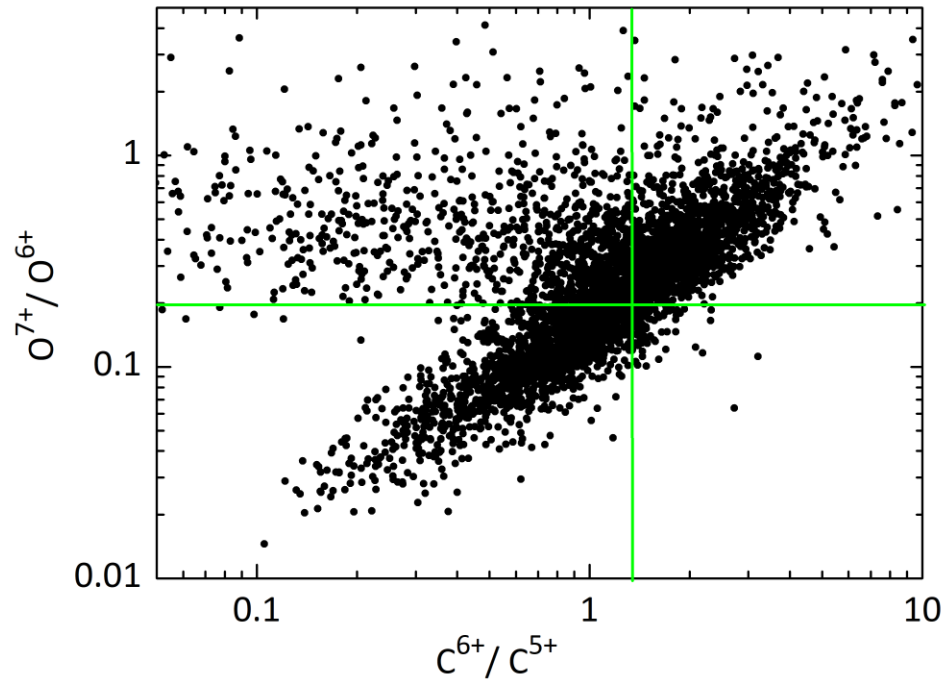


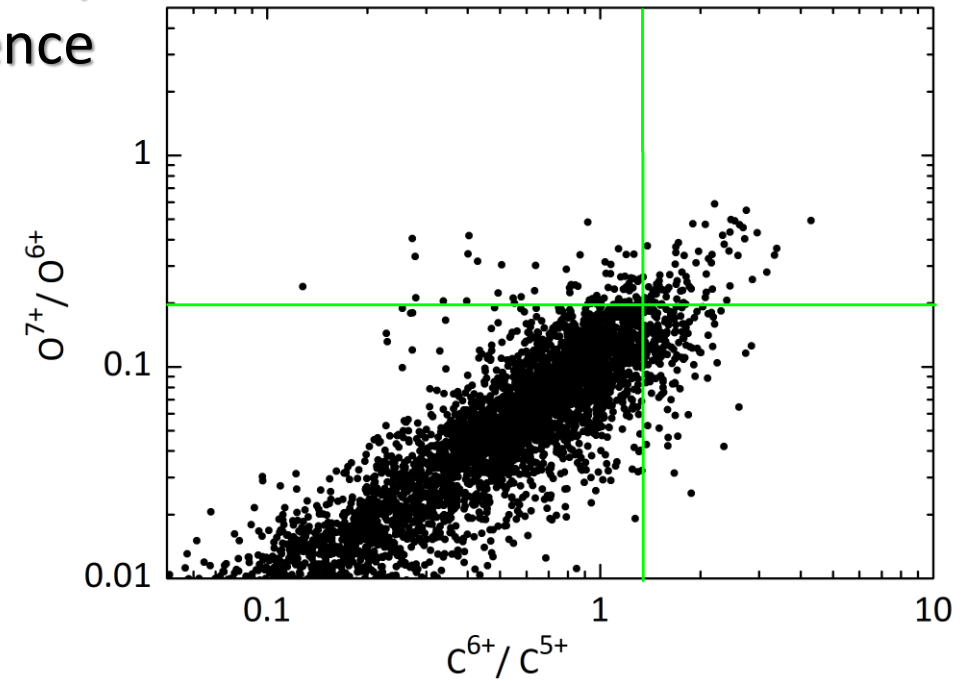
Figure 2. Scatter density plots of predicted [from linear regression model](#) versus observed ion composition ratios for three species: C^{6+}/C^{5+} (left), O^{7+}/O^{6+} (middle), and Fe/O (right) of Experiment 2 in Table 1. The red dashed line represents one-to-one line, and the Pearson correlation coefficient (CC) is shown in each panel.

Solar activity dependence

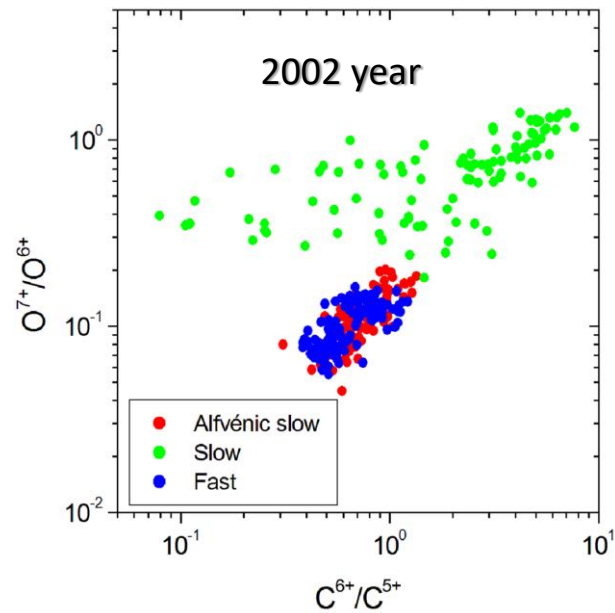
2001 Solar maximum



2008 Solar minimum



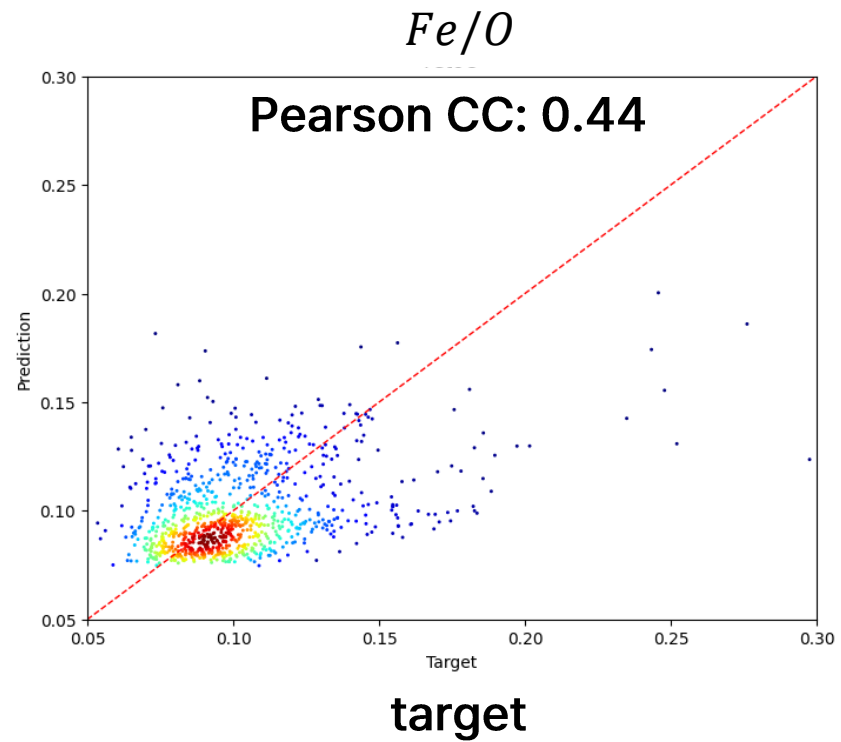
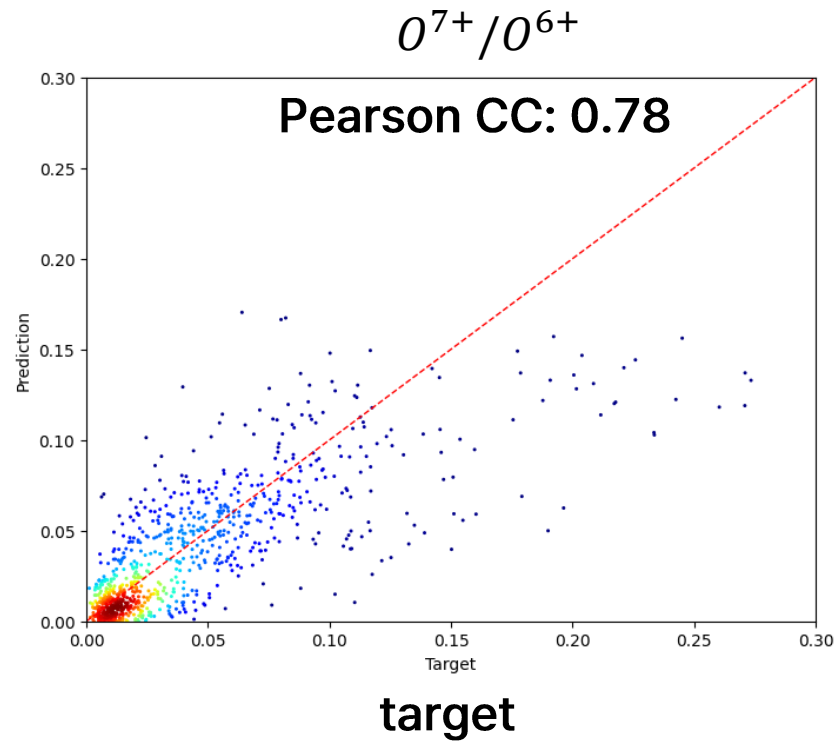
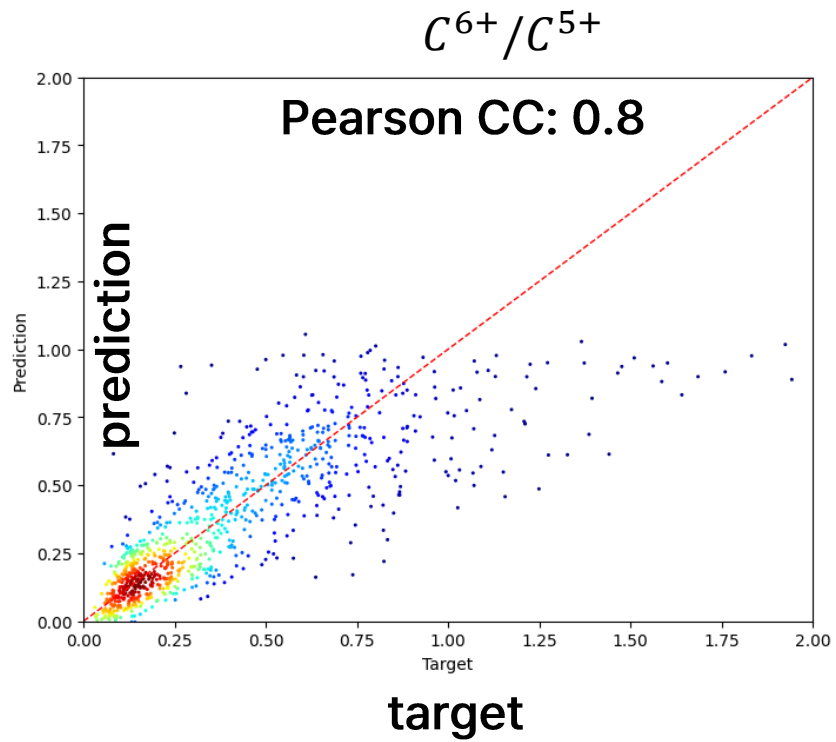
2002 year



- This indicates plasma coming from *different source regions*
- : a coronal-hole-associated solar wind - relatively *cold* source
 - : a streamer-associated slow wind - relatively *hot* source

D'Amicis + 2019

2007-2008 태양활동 극소기





Thank You

